Transformations & Symmetry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 9.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In ΔABC studying properties of rectangles, rhombuses, and squares, $m\angle A = 34^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 94°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (34^\circ + 52^\circ) = 94^\circ$.

2 **[Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

For the model $f(x) = 2x + 9$ in properties of trapezoids, evaluate $f(2)$.

Final Answer: 13. Substitute $x = 2$: $f(2) = 2(2) + 9 = 13$.

3 **[Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

A data set recorded for properties of trapezoids consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 17.5. The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

4 **[Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

[Transformations & Vectors · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

5 **[Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Find the magnitude of $\mathbf{v} = \begin{pmatrix} -5 \\ 12 \end{pmatrix}$ and hence find a unit vector in the direction of $\mathbf{v}$.

$|\mathbf{v}| = \sqrt{25 + 144} = \sqrt{169} = 13$. Unit vector $= \frac{1}{13}\begin{pmatrix} -5 \\ 12 \end{pmatrix}$.

6 **[Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

A price increased from 80 to 100 SAR. Find the percentage increase.

Step 1: Compute absolute change: $\Delta P = 100 - 80 = 20$ SAR. Step 2: Calculate percent increase relative to base price 80: $\text{Percent Increase} = \frac{\text{New} - \text{Old}}{\text{Old}} \times 100\% = \frac{20}{80} \times 100\% = \frac{1}{4} \times 100\% = \mathbf{25\%}$ The price increased by 25%.

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7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve the linear system $3x + 2y = 16$ and $x - y = 2$ for x and y.

Step 1: From the second equation, isolate x: $x = y + 2$ Step 2: Substitute into the first equation: $3(y + 2) + 2y = 16 \Rightarrow 3y + 6 + 2y = 16 \Rightarrow 5y + 6 = 16 \Rightarrow 5y = 10 \Rightarrow \mathbf{y = 2}$ Step 3: Substitute back to find x: $x = 2 + 2 = \mathbf{4}$ The unique solution is **(4, 2)**.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A car travels a distance of $d = 180$ km correct to the nearest 10 km in a time of $t = 2.4$ hours correct to 1 decimal place. Calculate the upper bound for the average speed of the car in km/h.

Find the bounds for distance and time: $d_{\text{upper}} = 180 + 5 = 185$ km $t_{\text{lower}} = 2.4 - 0.05 = 2.35$ hours To maximize speed $s = \frac{d}{t}$, divide the upper distance by the lower time: $s_{\text{upper}} = \frac{185}{2.35} \approx 78.723\ldots \approx 78.7$ km/h (3 s. f.)

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 3 · AdvancedA researcher randomly assigns 50 students to a tutoring programme and 50 to a control group, then compares test scores. Which conclusion is best supported if the tutored group scores significantly higher? (A) Tutoring causes higher scores for the population sampled. (B) There is an association, but causation cannot be determined. (C) Results apply to all high school students nationwide. (D) The sample size is too small to draw any valid conclusion.

Correct Answer: (A). Because participants were randomly assigned to the treatment and control groups (a randomized controlled experiment), a causal conclusion can be established. Because the sample was drawn from a specific student population, the result generalizes to that sampled population (not nationwide, eliminating C).

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A USD 80 item is marked down 20%. Find the sale price.

Percent Decrease Multiplier Shortcut: A markdown of 20% means the customer pays $100\% - 20\% = 80\%$ of the original price. Sale Price $= 80 \times (1 - 0.20) = 80 \times 0.80 =$ **USD64** Trap Warning: Do NOT subtract $USD20$ directly; 20% of $USD80$ is $USD16$, leaving $USD80 - USD16 = USD64$.

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11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The equation of a circle is given by $x^2 + y^2 + 10x - 4y + 20 = 0$. What is the radius of this circle?

Group terms and complete the square for x and y: $(x^2 + 10x) + (y^2 - 4y) = -20$ Half the coefficients squared: $(10/2)^2 = 25$, $(-4/2)^2 = 4$. Add $25 + 4 = 29$ to both sides: $(x^2 + 10x + 25) + (y^2 - 4y + 4) = -20 + 25 + 4 = 9$ $(x + 5)^2 + (y - 2)^2 = 9$ Since $r^2 = 9$, the radius is $r = \sqrt{9} = \mathbf{3}$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

If 5 workers finish a task in 12 days, how many days for 3 workers?

Step 1: Compute total worker-days needed: Since work is constant, workers and days vary inversely: $W = 5 \text{ workers} \times 12 \text{ days} = 60 \text{ worker} - \text{days}$ Step 2: Determine days for 3 workers: $t = \frac{60 \text{ worker} - \text{days}}{3 \text{ workers}} = \mathbf{20} \text{ days}$ It will take 20 days.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A shop offers a 15% discount on a 200 SAR item. Find the sale price.

Step 1: Determine the discounted multiplier: A 15% discount leaves $100\% - 15\% = 85\%$ (multiplier 0.85). Step 2: Calculate the sale price: $P_{\text{sale}} = 200 \times 0.85 = \mathbf{170}$ SAR (Alternatively: Discount is $200 \times 0.15 = 30$ SAR; sale price is $200 - 30 = 170$ SAR).

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 4 · SynthesisIn the system of equations $ax + 8y = 24$ and $6x + by = 8$, a and b are constants. If the system has infinitely many solutions, what is the value of $a + b$?

The constant ratio from the first equation to the second is $\frac{24}{8} = 3$. For infinitely many solutions, every coefficient in the first equation must be 3 times the corresponding coefficient in the second: $a = 3 \times 6 = 18, \quad 8 = 3b \Rightarrow b = \frac{8}{3}.\ a + b = 18 + \frac{8}{3} = \frac{62}{3}.$

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Triangles & Congruence

Level 4 · SynthesisLine ℓ passes through $(2, -1)$ and is perpendicular to the line $3x + 6y = 18$. If line ℓ also passes through $(k, 9)$, what is the value of k?

Rewrite $3x + 6y = 18$ in slope-intercept form: $6y = -3x + 18 \Rightarrow y = -\frac{1}{2}x + 3$, so its slope is $-\frac{1}{2}$. The perpendicular line ℓ has slope $m_\ell = 2$. Using $(2, -1)$ and $(k, 9)$: $\frac{9 - (-1)}{k - 2} = 2 \Rightarrow \frac{10}{k - 2} = 2 \Rightarrow k - 2 = 5 \Rightarrow k = 7.$

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16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

Solve the logarithmic equation $\log_3(x^2 - 7) = 2$ for all real solutions, and verify that all roots lie within the valid domain.

Step 1: Convert to exponential form: $x^2 - 7 = 3^2 = 9$. Step 2: Solve the quadratic equation: $x^2 = 9 + 7 = 16 \Rightarrow x = \pm 4$. Step 3: Verify the domain $x^2 - 7 > 0$: For both $x = 4$ and $x = -4$: $x^2 - 7 = 16 - 7 = 9 > 0$. Both roots are valid. The solutions are $\mathbf{x = 4}$ and $\mathbf{x = -4}$.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Right Triangles & Trigonometry

A line is parallel to $y = 2x + 1$. What is its slope?

Parallel lines in the coordinate plane have identical slopes ($m_1 = m_2$) and different y-intercepts. Slope of $y = 2x + 1$ is $m = 2 \Rightarrow m_{\parallel} = \mathbf{2}$ Contrast: A line perpendicular to this line would have a slope of $m_{\perp} = -\frac{1}{2}$ (negative reciprocal).

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Polygons & Quadrilaterals

Two angles of a triangle are 55° and 75°. Find the third.

Step 1: Use triangle angle sum theorem: The three interior angles of any triangle sum to 180°: $\theta + 55^\circ + 75^\circ = 180^\circ$ Step 2: Solve for θ: $\theta + 130^\circ = 180^\circ \Rightarrow \theta = 180^\circ - 130^\circ = \mathbf{50}^\circ$ The third angle measures $\mathbf{50}^\circ$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Circles

A car travels 120 miles in 3 hours. Find the average speed.

Apply the fundamental distance-rate-time relationship $d = r \cdot t$: $r = \frac{d}{t} = \frac{120 \text{ miles}}{3 \text{ hours}} = \mathbf{40}$ mph ACT Exam Note: Average speed is always Total Distance divided by Total Time. Never average two speeds directly if the travel times are different.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Area, Surface Area & Volume

Point $M(3, -2)$ is the midpoint of line segment $\overline{AB}$. If point A has coordinates $(-1, 4)$, what are the coordinates of point B?

Using the midpoint formula for each coordinate: $x_M = \frac{x_A + x_B}{2} \Rightarrow 3 = \frac{-1 + x_B}{2} \Rightarrow 6 = -1 + x_B \Rightarrow x_B = 7$. $y_M = \frac{y_A + y_B}{2} \Rightarrow -2 = \frac{4 + y_B}{2} \Rightarrow -4 = 4 + y_B \Rightarrow y_B = -8$. Therefore, the coordinates of B are (7, -8).

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying angles of polygons, $m\angle A = 32^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 103° ✓

b. 113°

c. 95°

d. 148°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (32^\circ + 45^\circ) = 103^\circ$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying properties of parallelograms, $m\angle A = 33^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 102° ✓

b. 112°

c. 94°

d. 147°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (33^\circ + 45^\circ) = 102^\circ$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for classifying quadrilaterals consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5 ✓

d. 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 3x + 3$ in classifying quadrilaterals, evaluate $f(4)$.

a. 15 ✓

b. 18

c. 12

d. 19

Substitute $x = 4$: $f(4) = 3(4) + 3 = 15$.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for coordinate methods with quadrilaterals consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 4x + 4$ in coordinate methods with quadrilaterals, evaluate $f(4)$.

a. 20 ✓

b. 24

c. 16

d. 24

Substitute $x = 4$: $f(4) = 4(4) + 4 = 20$.

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Under a translation by the vector $\langle 5, -3 \rangle$, the image of point P is $P'(2, 1)$. What are the coordinates of the pre-image point P?

a. $(-3, 4)$ ✓

b. $(7, -2)$

c. $(-3, -2)$

d. $(7, 4)$

A translation by the vector $\langle a, b \rangle = \langle 5, -3 \rangle$ maps a pre-image point $P(x, y)$ to its image $P'(x', y')$ according to the rule: $(x', y') = (x + a, y + b) = (x + 5, y - 3)$ We are given the coordinates of the image $P'(2, 1)$, so we set up equations to recover the pre-image coordinates x and y:
$x + 5 = 2 \Rightarrow x = 2 - 5 = -3$ $y - 3 = 1 \Rightarrow y = 1 + 3 = 4$ Therefore, the pre-image point is $P(-3, 4)$.
Common error analysis: - Choice $(7, -2)$ results from incorrectly a.

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

What is the order of rotational symmetry of a regular decagon?

a. 20

b. 9

c. **10** ✓

d. 1

A regular n-gon has rotational symmetry of order n (rotations by $360^\circ/n$); here $n = 10$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

What is the order of rotational symmetry of a regular dodecagon?

a. 1

b. 24

c. 11

d. **12** ✓

A regular n-gon has rotational symmetry of order n (rotations by $360^\circ/n$); here $n = 12$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the order of rotational symmetry of an equilateral triangle?

a. 1

b. 2

c. **3** ✓

d. 6

A regular n-gon has rotational symmetry of order n (rotations by $360^\circ/n$); here $n = 3$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the order of rotational symmetry of a square?

a. 8

b. 1

c. **4** ✓

d. 3

A regular n-gon has rotational symmetry of order n (rotations by $360^\circ/n$); here $n = 4$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

What is the order of rotational symmetry of a regular pentagon?

a. 4

b. **5** ✓

c. 1

d. 10

A regular n-gon has rotational symmetry of order n (rotations by $360^\circ/n$); here $n = 5$.

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

What is the order of rotational symmetry of a regular hexagon?

a. 5

b. 12

c. **6** ✓

d. 1

A regular n-gon has rotational symmetry of order n (rotations by $360^\circ/n$); here $n = 6$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

What is the order of rotational symmetry of a regular heptagon?

a. 6

b. **7** ✓

c. 1

d. 14

A regular n-gon has rotational symmetry of order n (rotations by $360^\circ/n$); here $n = 7$.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

What is the order of rotational symmetry of a regular octagon?

a. **8** ✓

b. 7

c. 16

d. 1

A regular n-gon has rotational symmetry of order n (rotations by $360^\circ/n$); here $n = 8$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the order of rotational symmetry of a regular nonagon?

a. 18

b. 1

c. 8

d. **9** ✓

A regular n-gon has rotational symmetry of order n (rotations by $360^\circ/n$); here $n = 9$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Reflecting $(-2, 3)$ over the line $y = x$ gives:

a. (2,-3)

b. **(3,-2)** ✓

c. (-2,3)

d. (-3,2)

Reflection over the line $y = x$ swaps the coordinates: $(-2, 3) \to (3, -2)$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Polygons & Quadrilaterals

Reflecting $(3, -2)$ over x-axis gives:

a. (-2,3)

b. (-3,2)

c. **(3,2)** ✓

d. (3,-2)

Reflection over x-axis negates the y-coordinate: $(3, -2) \to (3, 2)$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Circles

Reflecting $(4, 5)$ over y-axis gives:

a. (-4,-5)

b. (4,5)

c. (5,4)

d. **(-4,5)** ✓

Reflection over y-axis negates the x-coordinate: $(4, 5) \to (-4, 5)$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Area, Surface Area & Volume

Reflect $(3, -2)$ over the x-axis, then translate by $\langle 1, 2 \rangle$. Find the final image.

a. **(4,4)** ✓

b. (4,0)

c. (-2,0)

d. (3,2)

Reflect: $(3, -2) \to (3, 2)$. Translate: $(3, 2) \to (4, 4)$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Foundations & Reasoning

How many lines of symmetry does a regular decagon have?

a. 9

b. 20

c. **10** ✓

d. 11

A regular n-gon has exactly n lines of symmetry; here $n = 10$.

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 10.01 Coordinate Geometry

How many lines of symmetry does a regular dodecagon have?

a. 11

b. **12** ✓

c. 13

d. 24

A regular n-gon has exactly n lines of symmetry; here $n = 12$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 11.01 Probability

How many lines of symmetry does an equilateral triangle have?

a. 2

b. 6

c. **3** ✓

d. 4

A regular n-gon has exactly n lines of symmetry; here $n = 3$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Lines & Angles

How many lines of symmetry does a square have?

a. 8

b. 3

c. **4** ✓

d. 5

A regular n-gon has exactly n lines of symmetry; here $n = 4$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Triangles & Congruence

How many lines of symmetry does a regular pentagon have?

a. 10

b. **5** ✓

c. 4

d. 6

A regular n-gon has exactly n lines of symmetry; here $n = 5$.

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